A B C A L C U L U S

CHAPTER 1

Limits & Continuity

AP-Style Practice Questions

EASY MEDIUM HARD

Topics **1.1 – 1.16** AB

Name: _____ Period: _____

PART I

Topics 1.1 – 1.16

Multiple Choice Questions

Show all supporting work on scratch paper. On the AP Exam, Section I is split into a no-calculator and a calculator-allowed part — each question below is labeled accordingly.

Q1 EASY 1.3 Estimating from Tables No Calculator

The table gives values of $f(x)$ near $x = 2$.

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$	4.61	4.9601	4.996	5.004	5.0401	5.41

Based on the table, $\lim_{x \to 2} f(x)$ is best estimated by

(A) 4

(B) 4.5

(C) 5

(D) does not exist

Q2 EASY 1.5 Algebraic Manipulation No Calculator

$$\lim_{x \to 3} \frac{x^2 - 9}{x - 3} =$$

(A) 0

(B) 3

(C) 6

(D) does not exist

Q3 EASY 1.6 Rationalizing No Calculator

$$\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4} =$$

(A) 0

(B) $\frac{1}{4}$

(C) $\frac{1}{2}$

(D) 1

Q4 MEDIUM 1.8 Special Trig Limits No Calculator

$$\lim_{x \to 0} \frac{\sin(5x)}{3x} =$$

(A) 0

(B) $\frac{3}{5}$

(C) $\frac{5}{3}$

(D) does not exist

Q5 MEDIUM 1.8 Trig Limits No Calculator

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2} =$$

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) does not exist

Q6 MEDIUM 1.7 Squeeze Theorem No Calculator

If $4 - x^2 \leq g(x) \leq 4 + x^2$ for all x, then $\lim_{x \to 0} g(x) =$

(A) 0

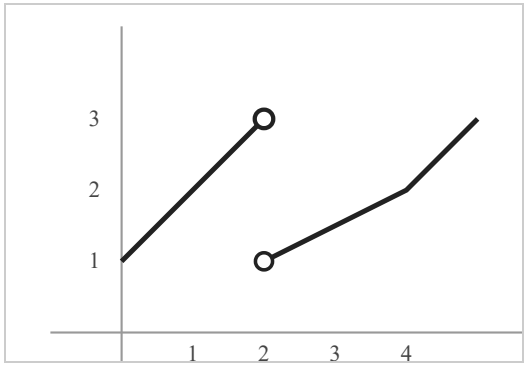
(B) 2

(C) 4

(D) cannot be determined

Q7 EASY 1.2 One-Sided from Graph No Calculator

The graph of f is shown. Which statement is true?



(A) $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x)$ and $f(2)$ equals that value.

(B) $\lim_{x\to 2^-} f(x) \neq \lim_{x\to 2^+} f(x)$, so $\lim_{x\to 2} f(x)$ does not exist.

(C) $\lim_{x\to 2} f(x)$ exists and equals $f(2)$.

(D) $\lim_{x\to 2} f(x)$ exists but is not equal to $f(2)$.

Q8 MEDIUM 1.9 Continuity No Calculator

Let $f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x \neq 2 \\ k, & x = 2 \end{cases}$. For what value of k is f continuous at $x = 2$?

(A) 0

(B) 2

(C) 4

(D) no such value exists

Q9 MEDIUM 1.10 Types of Discontinuity No Calculator

$g(x) = \dfrac{x+1}{x^2+x}$ has which type of discontinuity at $x = 0$?

(A) Removable

(B) Jump

(C) Infinite

(D) g is continuous at $x = 0$

Q10 HARD 1.11 Piecewise Continuity No Calculator

Find a and b so that $f(x) = \begin{cases} 2x + a, & x \le 1 \\ bx^2 + 3, & 1 < x < 2 \\ 4x - b, & x \ge 2 \end{cases}$ is continuous everywhere.

(A) $a = 2,\ b = 1$

(B) $a = 3,\ b = 2$

(C) $a = 1,\ b = 0$

(D) No such (a, b) makes f continuous everywhere.

Q11 MEDIUM 1.14 Infinite Limits No Calculator

$$\lim_{x \to 2^-} \frac{x+3}{x-2} =$$

(A) $+\infty$

(B) $-\infty$

(C) 0

(D) 1

Q12 MEDIUM 1.15 Limits at Infinity No Calculator

$$\lim_{x \to \infty} \frac{6x^2 - x}{3x^2 + 4} =$$

(A) 0

(B) 1

(C) 2

(D) ∞

Q13 HARD 1.15 Limits at Infinity (Radical) No Calculator

$$\lim_{x\to\infty} \frac{\sqrt{9x^4+1}}{x^2-3x} =$$

(A) 0

(B) 1

(C) 3

(D) nonexistent

Q14 HARD 1.15 End Behavior No Calculator

$$\lim_{x\to-\infty} \left(\sqrt{x^2+4x}+x\right) =$$

(A) -2

(B) 0

(C) 2

(D) ∞

Q15 MEDIUM 1.16 IVT No Calculator

Let f be continuous on $[0,3]$ with $f(0)=-2$ and $f(3)=5$. Which conclusion does the IVT guarantee?

(A) $f(c)=0$ for some $c\in(0,3)$.

(B) $f(c)=6$ for some $c\in(0,3)$.

(C) $f(c)=-3$ for some $c\in(0,3)$.

(D) f is differentiable on $(0,3)$.

Q16 HARD 1.16 IVT (Table) No Calculator

The continuous function h has the selected values below. What is the minimum number of real zeros of h on $[1,9]$ guaranteed by the IVT?

x	1	3	5	7	9
$h(x)$	-4	2	-1	3	-5

(A) 1

(B) 2

(C) 3

(D) 4

Q17 MEDIUM 1.12 Intermediate Forms No Calculator

$$\lim_{h\to 0} \frac{(2+h)^3 - 8}{h} =$$

(A) 0

(B) 6

(C) 8

(D) 12

Q18 HARD 1.13 Complex Fractions No Calculator

$$\lim_{x\to 0} \frac{\frac{1}{x+3} - \frac{1}{3}}{x} =$$

(A) $-\frac{1}{9}$

(B) 0

(C) $\frac{1}{9}$

(D) does not exist

PART II *Show All Work*

Free-Response Questions

Free-response answers must include complete setup: algebraic manipulation, stated theorem conditions for IVT/Squeeze, and interval justification for continuity. Units and contextual explanations are required where indicated.

FRQ 1 EASY 1.5 / 1.6 Evaluating Limits No Calculator

Evaluate each limit. Show all algebraic steps.

(a) $\lim_{x \to 5} \dfrac{x^2 - 25}{x^2 - 4x - 5}$

(b) $\lim_{x \to 9} \dfrac{x - 9}{\sqrt{x} - 3}$

(c) $\lim_{x \to 0} \dfrac{\sin(3x)}{\tan(2x)}$

FRQ 2 MEDIUM 1.9 – 1.11 Continuity & Parameters No Calculator

Let $f(x) = \begin{cases} \dfrac{x^2 - x - 6}{x - 3}, & x < 3 \\ ax + b, & 3 \le x \le 5 \\ x^2 - 9, & x > 5 \end{cases}$.

(a) Find $\lim_{x \to 3^-} f(x)$ and explain how this determines a restriction on a and b.

(b) Determine values of a and b that make f continuous on $\mathbb{R}$. Show the system you solve.

(c) With the values from (b), classify the discontinuity of f' at $x = 3$ and at $x = 5$ (if any). *(Preview of Unit 2 — derivative discontinuity. Cram-track students may skip this part.)*

FRQ 3 MEDIUM 1.14 – 1.15 Asymptotes No Calculator

Let $f(x) = \dfrac{2x^2 - x - 6}{x^2 - 4}$.

(a) Find all vertical asymptotes of f. Justify with one-sided limits.

(b) Find any removable discontinuities and state the value needed to remove each.

(c) Find $\lim\limits_{x \to \infty} f(x)$ and $\lim\limits_{x \to -\infty} f(x)$, and state the horizontal asymptote.

FRQ 4 HARD 1.16 IVT Application (Table) Calculator

A diver's depth $d(t)$ in meters at time t seconds is continuous on $[0, 20]$.

t (s)	0	4	10	15	20
$d(t)$ (m)	0	8	22	18	5

(a) Use the IVT to justify that there is a time in $(0, 10)$ when the diver is exactly 15 meters deep.

(b) Is the IVT enough to conclude that the diver is 15 meters deep at some time in $(10, 20)$? Justify.

(c) A student claims the IVT guarantees that the diver's depth equals 25 meters for some $t \in (0, 20)$. Is the claim correct? Explain.

(d) What is the minimum number of times the diver can be at depth 15 meters on $[0, 20]$? Justify.

FRQ 5 HARD 1.7 / 1.12 Squeeze & Definition No Calculator

Let $f(x) = x^2 \cos\left(\frac{1}{x}\right)$ for $x \neq 0$, and define $f(0) = 0$.

(a) Show, using the Squeeze Theorem, that $\lim_{x \to 0} f(x) = 0$. State the bounding inequalities and verify all conditions.

(b) Use part (a) to explain why f is continuous at $x = 0$.

(c) Compute $\lim_{x \to 0} \dfrac{f(x) - f(0)}{x - 0}$ using the Squeeze Theorem. Interpret the meaning of the limit.